

Q3. Explain the following compiler errors and why they might happen

1. “Stack Overflow”

This compiler error means the program used too much stack memory, this happens usually because of infinite recursion or by a function that creates very large local variables.

2. “Undefined reference to main”

This linker error means the linker cannot find the `main()` function. This happens because of missing `main()` function or when compiling multiple `.cpp` files, the developer did not include the file that contains `main()`.

3. “fooVar” undeclared (first use this function)

This compiler error means the compiler does not identify the variable `fooVar`. This happens because the variable was never declared or is out of scope or is misspelled.

4. `foo.h`: No such file or directory

This compiler error means the compiler cannot find that header file. This happens because of wrong filename, wrong file path, wrong include quotes/brackets or the file is not in the directory.

5. “Unterminated string or character constant”

This compiler error means a string or character literal was started but not properly closed. This happens mainly because of the missing closing quote of the string or character.

6. “ANSI C++ forbids comparison between pointer and integer”

This compiler error means the developer is comparing an address with an ordinary integer. This happens because the developer is mixing up a pointer and the value it points to.

7. “Null pointer exception” or “Segmentation Fault”

This runtime error means the program tried to access invalid memory address. This error happens because the developer was dereferencing a null pointer or dereferencing a dangling pointer or going out of bounds of an array.

8. “cout undeclared (first use this function)”

This compiler error means the compiler does not know what the object cout is. This error happens because the developer did not include `#include <iostream>`, did not use `using namespace std;`, and may forget `std::cout`.

9. “Parse error before something”

This compiler error means the compiler got confused by invalid syntax before that token. This error happens because of missing syntax like semicolon, bracket, parenthesis and wrong keyword.

10. “‘Member’ is not accessible”

This compiler error means the developer is trying to use a class member that is not public in that context. This error happens when the developer is trying to access a private or protected member from outside the class.

11. “Destructor name must match the class name”

This compiler error means that the destructor was declared with the wrong name. This happens because the destructor name must be the exactly same class name plus `~` at the front.

12. “Friends must be functions or classes”

This compiler error means the friend keyword was used on something invalid. This happens because only a function or a class can be declared as a friend, other things like variables cannot.